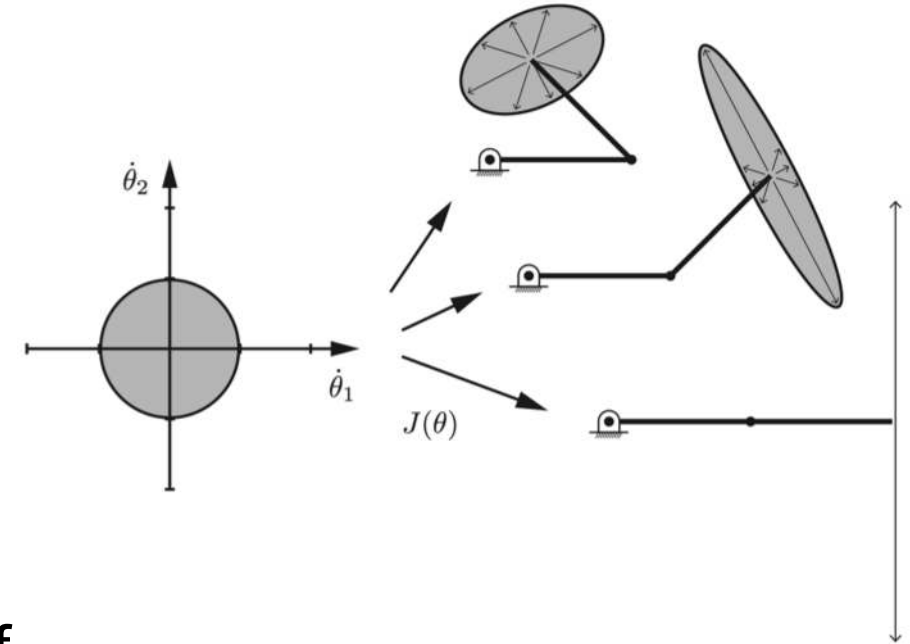


Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
	5.1 Manipulator Jacobian
	5.2 Statics of Open Chains
	5.3 Singularity Analysis
	5.4 Manipulability
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

- A configuration $\theta \in \mathbb{R}^n$ is **singular** if $\text{rank}(J(\theta))$ is less than its maximum rank over all θ .
- $\text{rank}(J_s(\theta)) = \text{rank}(J_b(\theta))$; singularities are independent of frame.
- Some common sources of singularities for six-dof spatial open chains:
 - i. two collinear revolute joint axes
 - ii. three coplanar and parallel revolute joint axes
 - iii. four revolute joint axes intersecting at a common point
 - iv. four coplanar revolute joints
 - v. six revolute joint axes intersecting a common line

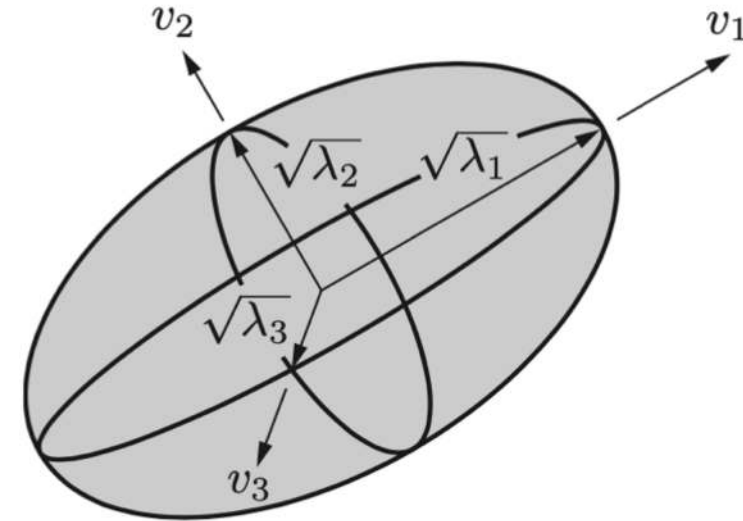


Important concepts, symbols, and equations (cont.)

- A **fat** or **wide** Jacobian (or matrix generally) has more columns than rows. A **skinny** or **tall** Jacobian has more rows than columns.
- A robot is **redundant** for a task if there exists a family of $\dot{\theta}$ satisfying $v = J(\theta) \dot{\theta}$ at most θ , where v could be a twist $\mathcal{V}$ or a coordinate velocity.
(wide Jacobians)
- A robot is **kinematically deficient** for a task if there are required velocities v for which there is no $\dot{\theta}$ satisfying $v = J(\theta) \dot{\theta}$ at most θ . (tall Jacobians)
- Redundancy and deficiency depend on the robot and the task (e.g., dimensions of the Jacobian); singularities depend on θ .

Important concepts, symbols, and equations (cont.)

- The **manipulability** of a robot at a configuration θ measures how close it is to being singular.
- The **manipulability ellipsoid** defines the ease of motion in different directions.
- The principal axes of the ellipsoid are given by the eigenvectors and square roots of the eigenvalues of the square matrix JJ^T .
- Manipulability measures include
 1. the ratio of the largest to smallest principal axis half-lengths
 2. the ratio of the largest to smallest eigenvalues
 3. the volume of the ellipsoid (proportional to $\sqrt{\lambda_1 \lambda_2 \dots}$).



Important concepts, symbols, and equations (cont.)

- Split the body Jacobian into angular and linear components to get angular velocity and linear velocity ellipsoids.

$$J_b(\theta) = \begin{bmatrix} J_{b\omega}(\theta) \\ J_{bv}(\theta) \end{bmatrix} \in \mathbb{R}^{6 \times n}$$

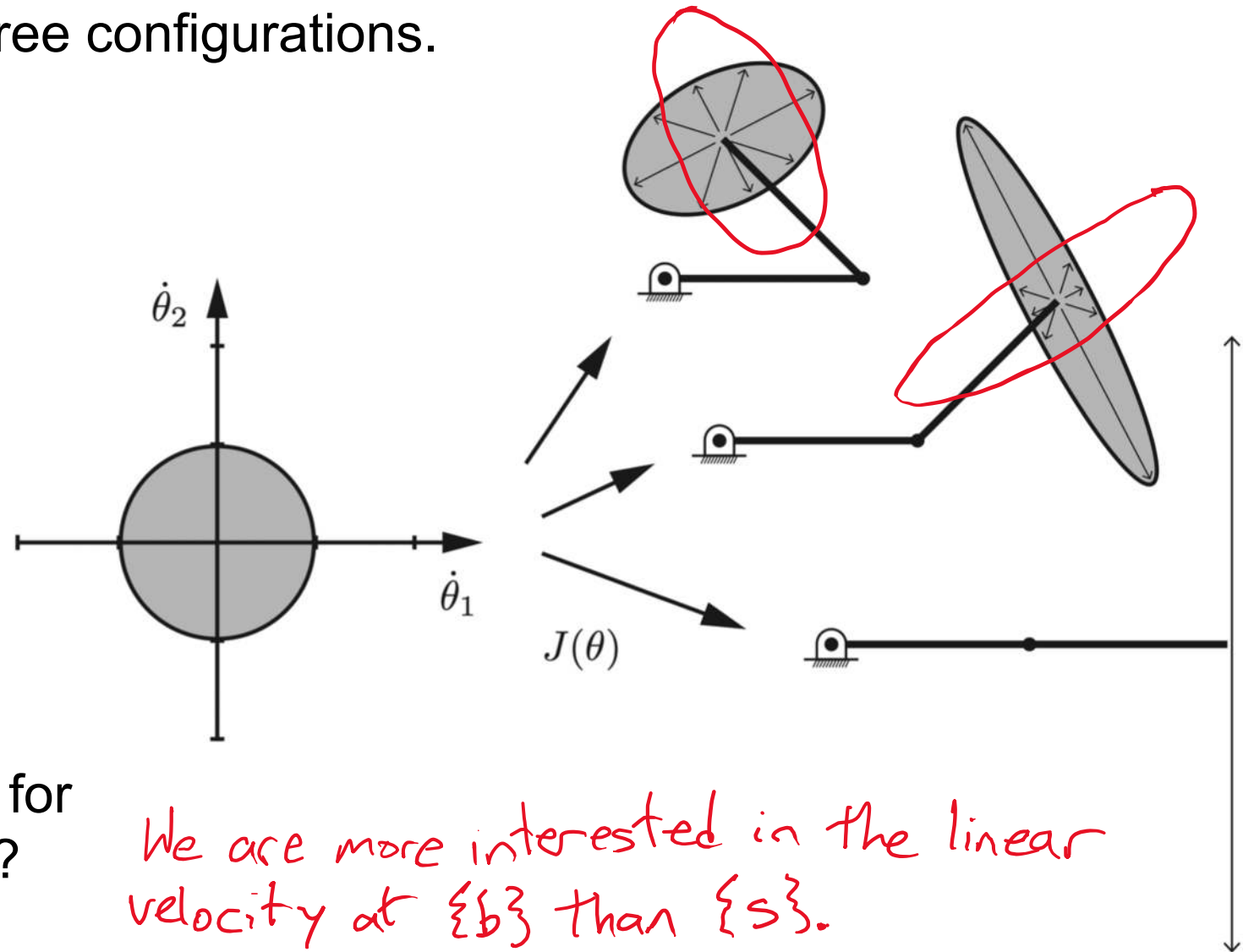
$J_{b\omega}(\theta) \in \mathbb{R}^{3 \times n} \rightarrow$ angular velocity/moment ellipsoids

$J_{bv}(\theta) \in \mathbb{R}^{3 \times n} \rightarrow$ linear velocity/force ellipsoids

Important concepts, symbols, and equations (cont.)

- The principal axes of the force (wrench) ellipsoid are given by the eigenvectors and square roots of the eigenvalues of $(JJ^T)^{-1}$.
- The force ellipsoid has the same principal axes as the manipulability ellipsoid, and the principal axis half-lengths are the reciprocal of those of the manipulability ellipsoid.
- The product of the volumes of the force and manipulability ellipsoids is constant over θ .

Approximately draw the corresponding force ellipsoids at the three configurations.



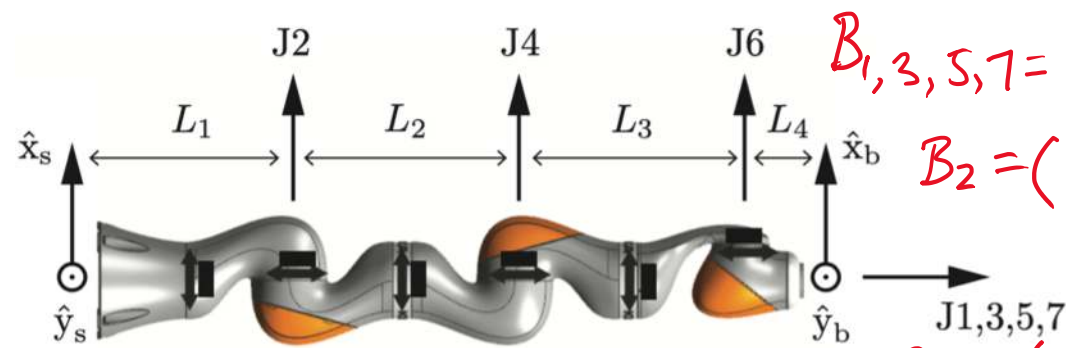
Why use J_b instead of J_s for manipulability ellipsoids?

We are more interested in the linear velocity at $\{b\}$ than $\{s\}$.



Adept SCARA RRRP robot

6×4
deficient



KUKA LBR iiwa 7R robot
 6×7 , redundant

$$B_{1,3,5,7} = (0, 0, 1, 0, 0, 0)$$

$$B_2 = (1, 0, 0, 0, -(L_2 + L_3 + L_4), 0)$$

$$B_4 = (1, 0, 0, 0, -L_3 - L_4, 0)$$

$$B_6 = (1, 0, 0, 0, -L_4, 0)$$

If using a twist to represent the e-e velocity, what are the dimensions of each Jacobian?

For the task of manipulating a rigid body, which of these is redundant or deficient?

What is the rank of the Jacobian of the iiwa at its home configuration (shown)?

$\text{rank}(J) = 3$ B_1, B_2, B_4 are linearly independent



UR5 6R robot
 6×6